Assessment Menu: Teacher Guide

The Reasoning Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
MEASURE	Segments, midpoint and distance	A1, A5, E1, E2, S1
ANGLES	Angle relationships with algebra	A2, E3, E4, S2
LOGIC	Conditionals, converse, inverse, contrapositive	A3, A6, E5, S3
PROOF	Counterexamples and proof reasoning	A4, E6, S4

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 MEASURE	Find the midpoint of the segment with endpoints A(−2, 5) and B(6, 1).	**Answer:** (2, 3) Average the x-coordinates: $\frac{-2+6}{2}$ = 2 Average the y-coordinates: $\frac{5+1}{2}$ = 3 Midpoint (2, 3)	2 pts: Averages the x-coordinates correctly 2 pts: Averages the y-coordinates correctly 1 pts: Writes the answer as an ordered pair (2, 3)
A2 ANGLES	Two angles form a linear pair. One measures (x + 20)° and the other measures 100°. Find x.	**Answer:** x = 60 A linear pair is supplementary: (x + 20) + 100 = 180 x + 120 = 180, so x = 60	2 pts: Writes (x + 20) + 100 = 180 1 pts: Names the reason: linear pair adds to 180° 2 pts: Correct x = 60
A3 LOGIC	Write the **converse** of: “If a number is divisible by 6, then it is even.” Is the converse true or false?	**Answer:** Converse: If a number is even, then it is divisible by 6. False (for example, 4). Swap the hypothesis and conclusion. Converse: If a number is even, then it is divisible by 6. False: 4 is even but is not divisible by 6.	2 pts: Swaps hypothesis and conclusion correctly 1 pts: Says false 2 pts: Gives a counterexample such as 4
A4 PROOF	Give a counterexample to the statement: “If $x^2 > 9$, then $x > 3$.”	**Answer:** $x = -4$: $x^2 = 16 > 9$, but -4 is not greater than 3. A counterexample makes the hypothesis true and the conclusion false. Try $x = -4$: $x^2 = 16$, which is greater than 9 (hypothesis true). But $-4 > 3$ is false (conclusion false).	2 pts: Chooses a valid value (any $x < -3$) 2 pts: Shows the hypothesis is true for it 1 pts: Shows the conclusion is false for it
A5 MEASURE	Find the distance between P(1, 2) and Q(4, 6).	**Answer:** 5 Use $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $d = \sqrt{3^2 + 4^2} = \sqrt{25}$ d = 5	2 pts: Differences 3 and 4 2 pts: Substitutes into the distance formula 1 pts: Answer 5
A6 LOGIC	Write the **contrapositive** of: “If a figure is a square, then it is a rectangle.”	**Answer:** If a figure is not a rectangle, then it is not a square. The contrapositive switches hypothesis and conclusion **and** negates both. If a figure is not a rectangle, then it is not a square.	2 pts: Swaps hypothesis and conclusion 2 pts: Negates both parts 1 pts: Complete if-then sentence

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 MEASURE	M(1, −2) is the midpoint of segment AB, and A is (−3, 4). Find the coordinates of B and show how you know.	**Answer:** B(5, −8) Midpoint x: $\frac{-3 + x}{2} = 1$, so −3 + x = 2 and x = 5 Midpoint y: $\frac{4 + y}{2} = -2$, so 4 + y = −4 and y = −8 B = (5, −8)	3 pts: Sets up the x midpoint equation 3 pts: Sets up the y midpoint equation 2 pts: Solves x = 5 2 pts: Solves y = −8
E2 MEASURE	Points A, B, C are collinear with B between A and C. AB = 2x + 3, BC = 4x − 5, and AC = 40. Find x, AB, and BC. Is B the midpoint of AC? Explain.	**Answer:** x = 7, AB = 17, BC = 23. B is not the midpoint because AB ≠ BC. Segment addition: (2x + 3) + (4x − 5) = 40 6x − 2 = 40, so x = 7 AB = 2(7) + 3 = 17 and BC = 4(7) − 5 = 23 (17 + 23 = 40 ✓) A midpoint needs AB = BC, but 17 ≠ 23.	3 pts: Segment addition equation 2 pts: x = 7 2 pts: AB = 17 and BC = 23 3 pts: Not the midpoint, since AB and BC differ
E3 ANGLES	Two lines intersect. One pair of vertical angles measures (4x − 7)° and (2x + 19)°. Find x, the measure of these angles, and the measure of an angle adjacent to one of them.	**Answer:** x = 13; the angles are 45°; an adjacent angle is 135°. Vertical angles are congruent: 4x − 7 = 2x + 19 2x = 26, so x = 13 4(13) − 7 = 45° Adjacent angles form a linear pair: 180 − 45 = 135°	3 pts: Sets the vertical angles equal 2 pts: x = 13 2 pts: Angle measure 45° 3 pts: Adjacent angle 135° using a linear pair
E4 ANGLES	Ray BD bisects ∠ABC. m∠ABD = (3x + 5)° and m∠DBC = (5x − 11)°. Find x and m∠ABC.	**Answer:** x = 8; m∠ABC = 58° A bisector makes two equal angles: 3x + 5 = 5x − 11 16 = 2x, so x = 8 m∠ABD = 3(8) + 5 = 29° and m∠DBC = 5(8) − 11 = 29° m∠ABC = 29 + 29 = 58°	3 pts: Sets the two parts equal (bisector) 2 pts: x = 8 2 pts: Finds each part, 29° 3 pts: m∠ABC = 58° (the whole angle, not just half)
E5 LOGIC	Statement: “If a figure is a rhombus, then it is a parallelogram.” Write its converse, inverse, and contrapositive. Decide whether each of the four is true or false, and give a counterexample for each false one.	**Answer:** Statement T; converse F (rectangle that is not a rhombus); inverse F (same figure); contrapositive T. Converse: If a figure is a parallelogram, then it is a rhombus. **False**: a 2 by 4 rectangle. Inverse: If a figure is not a rhombus, then it is not a parallelogram. **False**: the same rectangle. Contrapositive: If a figure is not a parallelogram, then it is not a rhombus. **True**. The original is **true**.	3 pts: Writes converse and inverse correctly 2 pts: Writes contrapositive correctly 2 pts: Correct truth value for all four 3 pts: Valid counterexamples for the converse and the inverse
E6 PROOF	Given: ∠1 and ∠2 are supplementary, and ∠2 and ∠3 are supplementary.	**Answer:** m∠1 = m∠3 by the Subtraction Property of Equality. 1. m∠1 + m∠2 = 180, m∠2 + m∠3 = 180 (def. of supplementary)	3 pts: Step 1, reason: def. of supplementary 3 pts: Step 2, reason: substitution

	Prove m∠1 = m∠3. Give a reason for each step.	2. m∠1 + m∠2 = m∠2 + m∠3 (substitution) 3. m∠1 = m∠3 (subtraction property)	3 pts: Step 3, reason: subtraction property 1 pts: Logical order

Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 MEASURE	Segment AB has midpoint M(2, 1) and endpoint A(−2, −3). **(a)** Find B. **(b)** Find AB exactly as a simplified radical. **(c)** Let C = (6, −3). Show that triangle ABC is isosceles.	**Answer:** (a) B(6, 5) (b) AB = $8\sqrt{2}$ (c) AC = BC = 8 (a) $\frac{-2 + x}{2} = 2$ gives x = 6; $\frac{-3 + y}{2} = 1$ gives y = 5. **B(6, 5)** (b) Differences are 8 and 8: AB = $\sqrt{(8^2 + 8^2)} = \sqrt{128} = 8\sqrt{2}$ (c) AC = 8 (same y, x differs by 8); BC = 8 (same x, y differs by 8). AC = BC, so isosceles.	3 pts: (a) x-coordinate 6 3 pts: (a) y-coordinate 5 3 pts: (b) Substitutes into the distance formula 3 pts: (b) Simplified radical 4 pts: (c) Finds AC = 8 4 pts: (c) Finds BC = 8 and concludes isosceles
S2 ANGLES	Lines AC and BD intersect at O. m∠AOB = (4x + 10)° and m∠BOC = (2x + 20)°. **(a)** Find x. **(b)** Find all four angles at O and name the reason for each. **(c)** Ray OE bisects ∠BOC. Find m∠AOE.	**Answer:** (a) x = 25 (b) ∠AOB = ∠COD = 110°, ∠BOC = ∠AOD = 70° (c) 145° (a) ∠AOB and ∠BOC are a linear pair: (4x + 10) + (2x + 20) = 180, so 6x = 150 and x = 25 (b) ∠AOB = 110° and ∠BOC = 70°. Vertical angles are congruent: ∠COD = 110° and ∠AOD = 70°. (c) OE splits ∠BOC into two 35° angles. ∠AOE = ∠AOB + 35° = 145°	3 pts: (a) Linear pair equation 2 pts: (a) x = 25 3 pts: (b) ∠AOB = 110°, ∠BOC = 70° 3 pts: (b) Vertical angles 1 pts: (b) Reasons named 3 pts: (c) Halves are 35° 3 pts: (c) ∠AOE = 145° 2 pts: Work organized
S3 LOGIC	True statements: “If a team wins, then it advances.” “If a team advances, then it plays in the finals.” **(a)** Combine them into one if-then statement and name the law. **(b)** The Tigers did not play in the finals. What follows? **(c)** Rosa says, “The Lions played in the finals, so they won.” Explain her error.	**Answer:** (a) If a team wins, then it plays in the finals (Law of Syllogism). (b) The Tigers did not win. (c) She assumed the converse. (a) Chain the statements: **If a team wins, then it plays in the finals.** This is the Law of Syllogism. (b) The contrapositive is true: if a team does not play in the finals, then it did not win. So the Tigers did not win. (c) Rosa used the converse (“if it plays in the finals, then it won”), which is not guaranteed: a team might reach the finals another way.	3 pts: (a) Combined statement 2 pts: (a) Law of Syllogism 3 pts: (b) Tigers did not win 4 pts: (b) Uses the contrapositive 4 pts: (c) Names the converse error 4 pts: (c) Explains why it fails
S4 PROOF	Given: A, B, C are collinear with B between A and C. M is the midpoint of AB and N is the midpoint of BC. Prove: MN = $\frac{1}{2}$AC. Write a proof with a reason for every step.	**Answer:** MN = $\frac{1}{2}$AC 1. MB = $\frac{1}{2}$AB, BN = $\frac{1}{2}$BC (def. of midpoint) 2. MN = MB + BN (segment addition) 3. MN = $\frac{1}{2}$AB + $\frac{1}{2}$BC (substitution) 4. MN = $\frac{1}{2}$(AB + BC) (distributive) 5. AB + BC = AC (segment addition) 6. MN = $\frac{1}{2}$AC (substitution)	4 pts: Def. of midpoint 4 pts: MN = MB + BN, segment addition 4 pts: Substitutes, factors out $\frac{1}{2}$ 4 pts: AB + BC = AC, segment addition 4 pts: Concludes; logical order, all reasons